

Sheet 1: Datasets

Exercise 1 (Jupyter Setup)

[2 Points]

This exercise is all about getting your working environment for the course up and running. Set up an installation of Jupyter that you will work with for the rest of the course. Your working environment must include proper dependency management such as `pipenv`, `conda` or `uv`. This contributes to the reproducibility of your work and facilitates easier collaboration within teams. A simple test of reproducibility is to clone/download your assignment into a fresh environment and try to run it. Optionally, though highly recommended, you can also have a look at additional tools such as `nbval` for validating your notebooks or `nbQA` for further quality assurance tools (code formatting, type annotations, static code analyses, etc.).

Exercise 2 (Dataset Wrangling)

[5 Points]

In this exercise, you will work with your first dataset. It is loosely defined to allow you to freely explore the datasets, coding environment, and libraries. We will learn more structured approaches towards dataset analyses later on in this course. We strongly recommend using this exercise to familiarize yourself with `pandas` or `polars` for analyzing and manipulating structured data.

- a) 1 Point Choose any dataset of your liking from e.g., [Kaggle](#), [Hugging Face](#), [data.gov](#), [UC Irvine Machine Learning Repository](#), [FiveThirtyEight](#), or [data.gv.at](#).
- b) 1 Point Read the data into a suitable data structure and manually inspect the data contained (columns, data formats, etc.).
- c) 2 Points Explore the dataset's characteristics (distributions, aggregations, plots, etc.).
- d) 1 Point Reflect on the metadata that you were presented for the dataset. Were the contents, formats, origin, etc. clear or was information missing?

Exercise 3 (Interactive Visualization)

[3 Points]

In Exercise 2, you already obtained a dataset and created some static visualizations. Now you are going to extend this and add *interactive* visualizations that allow the user of your notebook to easily explore the dataset. You can choose any library such as the popular `Plotly (Dash)` library, or `Perspective` or `Bokeh`.

Build a dashboard-style visualization of the dataset used in Exercise 2. Provide the user with interactive widgets to control what the dashboard shows. Your main goal should be to (i) familiarize

yourself with the library that you have chosen, so that you can use it effectively for future exercises and to (ii) explore and discuss the characteristics of the chosen dataset.

Submission: Submit the `.ipynb` and an exported `.pdf` or `.html` version of the notebook to OLAT.

Important: Submit your solution to OLAT and mark your solved exercises with the provided checkboxes. The deadline ends at 23:59 on the day before the discussion.